

AI Data Classification using K-Nearest Neighbors (KNN)

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AI Domain

June 12, 2026

Abstract

Artificial Intelligence and Machine Learning have become essential technologies for solving real-world classification problems. This project focuses on developing a supervised machine learning model capable of classifying Iris flower species based on their physical characteristics. The K-Nearest Neighbors (KNN) algorithm is employed to perform classification by analyzing similarities between data points. The project includes data preprocessing, feature scaling, model training, prediction, and performance evaluation using Accuracy Score, F1-Score, Classification Report, and Confusion Matrix. The developed system demonstrates the fundamental workflow of supervised learning and provides a strong foundation for advanced machine learning applications.

1 Introduction

Machine Learning is a branch of Artificial Intelligence that enables systems to learn patterns from data and make predictions without explicit programming. Classification is one of the most common supervised learning tasks where the objective is to assign data points to predefined categories.

In this project, the Iris dataset is used to train a machine learning model capable of identifying flower species based on sepal and petal measurements. The K-Nearest Neighbors algorithm was selected due to its simplicity, effectiveness, and suitability for introductory classification tasks.

2 Problem Statement

Traditional classification tasks require manual analysis and decision-making, which can be time-consuming and error-prone. The objective of this project is to automate the classification process by developing a machine learning model that can accurately predict flower species based on numerical features.

3 Objectives

- To understand supervised machine learning concepts.
- To load and analyze a dataset.
- To perform data preprocessing and feature scaling.
- To split the dataset into training and testing sets.
- To train a K-Nearest Neighbors classification model.
- To evaluate model performance using standard metrics.
- To visualize classification results using a confusion matrix.

4 Technologies Used

4.1 Programming Language

Python

4.2 Libraries

NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn

4.3 Development Environment

Visual Studio Code, Git, GitHub

5 Dataset Description

The Iris dataset is a widely used benchmark dataset in machine learning. It consists of 150 samples belonging to three different flower species.

5.1 Features

1. Sepal Length
2. Sepal Width
3. Petal Length
4. Petal Width

5.2 Target Classes

- Iris Setosa
- Iris Versicolor
- Iris Virginica

6 System Architecture

Dataset → Data Preprocessing → Feature Scaling → Train-Test Split → KNN Model
Training → Prediction → Performance Evaluation → Result Visualization

7 Methodology

7.1 Step 1: Data Collection

The Iris dataset is loaded using the Scikit-Learn dataset module.

7.2 Step 2: Data Preprocessing

The dataset is examined and prepared for training. Input features and target labels are separated.

7.3 Step 3: Feature Scaling

StandardScaler is applied to normalize feature values and improve model performance.

7.4 Step 4: Train-Test Split

The dataset is divided into training and testing subsets using an 80:20 ratio.

7.5 Step 5: Model Training

The K-Nearest Neighbors algorithm is trained using the processed training data.

7.6 Step 6: Prediction

The trained model predicts class labels for unseen testing data.

7.7 Step 7: Evaluation

Performance metrics such as Accuracy Score, F1-Score, Classification Report, and Confusion Matrix are generated.

8 Algorithm Used

8.1 K-Nearest Neighbors (KNN)

KNN is a supervised machine learning algorithm used for classification and regression tasks. It classifies a data point by analyzing the classes of its nearest neighboring data points.

8.2 Advantages

- Easy to implement
- No training complexity
- Effective for small datasets
- High accuracy for structured data

8.3 Limitations

- Computationally expensive for large datasets
- Sensitive to feature scaling
- Performance decreases with high-dimensional data

9 Results

The developed classification model achieved excellent performance on the testing dataset.

9.1 Evaluation Metrics

Accuracy Score, F1 Score, Classification Report, Confusion Matrix

The model successfully classified all testing samples with high prediction accuracy.

10 Test Cases

- **Test Case 1:** Input: Iris flower measurements. Expected Result: Correct species prediction. Status: Pass
- **Test Case 2:** Input: Preprocessed scaled data. Expected Result: Proper normalization. Status: Pass
- **Test Case 3:** Input: Testing dataset. Expected Result: Accurate classification. Status: Pass
- **Test Case 4:** Input: Model evaluation. Expected Result: Performance metrics generated. Status: Pass

11 Future Enhancements

- Implementation of advanced classification algorithms.
- Hyperparameter optimization.
- Support for larger datasets.
- Deployment as a web application.
- Real-time prediction interface.
- Comparison of multiple machine learning models.

12 Conclusion

This project successfully demonstrates the implementation of a supervised machine learning classification system using the K-Nearest Neighbors algorithm. The model was trained, tested, and evaluated using the Iris dataset and achieved excellent classification performance. The project provides practical exposure to data preprocessing, model training, prediction, and evaluation techniques, serving as a strong foundation for future machine learning applications.